



OLD DOMINION
UNIVERSITY



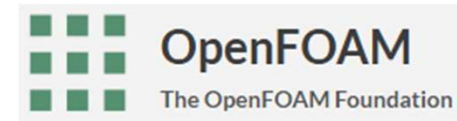
CCL Group Training

OpenFOAM Tutorial

Date 09/25/2026

OpenFOAM (ESI)

Open▽FOAM®

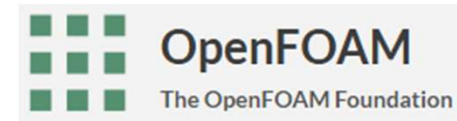


The ESI version of OpenFOAM (hosted at openfoam.com) is one of the two major release lines of the open-source CFD software. It is developed and maintained by OpenCFD Ltd., which is owned by ESI Group.

- **Historical Context:** OpenFOAM was originally released in 2004 by OpenCFD Ltd. In 2012, ESI Group acquired OpenCFD Ltd. Shortly after, the original developers split to form CFD Direct (which maintains openfoam.org), while ESI retained OpenCFD Ltd to develop openfoam.com.
- **Core Difference:** Both versions share the same core solvers (like [simpleFoam](#), [pimpleFoam](#), [blockMesh](#)) and basic syntax. However, the ESI version integrates features tailored for commercial workflows, including third-party software plugins and enterprise GUI support.

OpenFOAM (ESI)

OpenFOAM®



Upon installation, OpenFOAM includes a wide array of pre-compiled solvers categorized by physical domain.

Key Solver Categories & Primary Applications

Category	Representative Solvers	Primary Application / Use Case
Incompressible Flow	icoFoam, pisoFoam, pimpleFoam, simpleFoam	Subsonic Newtonian fluids where density changes are negligible (e.g., pipes, external aerodynamics).
Compressible Flow	rhoSimpleFoam, rhoPimpleFoam, rhoCentralFoam, sonicFoam	High-speed flows, acoustics, and shock waves ($Ma > 0.3$).
Multiphase Flow	interFoam, multiphaseEulerFoam, cavitatingFoam	Immiscible fluids, free-surface liquid/gas interfaces, and phase-change phenomena.
Heat Transfer & Combustion	buoyantSimpleFoam, chtMultiRegionFoam, reactingFoam	Buoyancy-driven thermal flows, conjugated heat transfer (fluid-solid), and chemical reaction models.
Particle & Dispersed Systems	DPMFoam, MPPICFoam, sprayFoam	Discrete Phase Model (DPM) particle tracking, sprays, and fluidized beds.
Electromagnetics & Solid Mechanics	electrostaticFoam, solidDisplacementFoam	Solid stress analysis, magnetohydrodynamics (MHD), and electrostatics.



OpenFOAM - Installation

1. Add the OpenFOAM.com Security Key

Download and store the security key required to authenticate packages from the OpenCFD repository:

```
curl -s https://dl.openfoam.com/add-debian-repo.sh | sudo bash
```

```
s3kim@ODU-JRZWF3:~$ curl -s https://dl.openfoam.com/add-debian-repo.sh | sudo bash
[sudo: authenticate] Password:
Detected distribution code-name: resolute
Added /etc/apt/sources.list.d/openfoam.list
Importing openfoam gpg key... done
Added /etc/apt/trusted.gpg.d/openfoam.gpg
Running apt-get update... done

The repository is setup! You can now install packages.
```



OpenFOAM - Installation

2. Install OpenFOAM

Update package listings and install the latest pre-compiled OpenFOAM release (openfoam-default points to the most recent version, such as openfoam2606 or newer):

```
sudo apt update
```

```
sudo apt install -y openfoam-default
```

```
s3kim@ODU-JRZWDF3:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://sourceforge.net/projects/openfoam/files/repos/deb resolute InRelease
15 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Some sources can be modernized. Run 'apt modernize-sources' to do so.
s3kim@ODU-JRZWDF3:~$ sudo apt install -y openfoam-default
Upgrading:
  curl  libcurl3t64-gnutls  libcurl4t64

Installing:
  openfoam-default

Installing dependencies:
  autoconf          krb5-multidev      libboost1.90-dev
  automake          libaec-dev         libbrotli-dev
  autotools-dev     libaec0            libc-dev-bin
  comerr-dev        libamd-comgr3      libc6-dev
  cpp               libamdhip64-7      libcaf-openmpi-3t64
```



OpenFOAM - Installation

3. Configure Environment Variables

To ensure the OpenFOAM shell commands are accessible automatically every time you open terminal, append the initialization script to your ~/.bashrc:

```
echo "source /usr/lib/openfoam/openfoam/etc/bashrc" >> ~/.bashrc  
source ~/.bashrc
```

Note:

- /usr/lib/openfoam/openfoam/etc/bashrc is a symbolic link pointing to the latest version installed (openfoam2606). Using this link ensures your configuration dynamically updates to the newest version whenever you upgrade OpenFOAM in the future.
- /usr/lib/openfoam/openfoam2606/etc/bashrc locks your configuration specifically to version 2606.

4. Verify the Installation

Confirm that your terminal recognizes the new environment:

```
simpleFoam -help
```

```
s3kim@ODU-JRZWDF3:~$ simpleFoam -help  
Usage: simpleFoam [OPTIONS]  
Options:  
-case <dir>          Case directory (instead of current directory)  
-decomposeParDict <file> Alternative decomposePar dictionary  
-dry-run             Check case set-up only using a simulation  
-dry-run-write       Check case set-up and write only  
-parallel            Run in parallel  
-postProcess         Execute functionObjects only  
-doc                 Display documentation in browser
```



OpenFOAM - Installation

5. Quick Test Simulation

Run a standard tutorial case to ensure the solvers work properly:

```
mkdir -p $FOAM_RUN
```

```
cd $FOAM_RUN
```

```
cp -r $FOAM_TUTORIALS/incompressible/simpleFoam/pitzDaily ./
```

```
cd pitzDaily
```

```
blockMesh
```

```
simpleFoam
```

```
s3kim@ODU-JRZWDF3:~/OpenFOAM/s3kim-v2606/run/pitzDaily$ simpleFoam
/*-----*
|=====|
| \      | F ield      | OpenFOAM: The Open Source CFD Toolbox
|  \     | O peration  | Version: 2606
|   \    | A nd        | Website: www.openfoam.com
|    \   | M anipulation|
|=====|
/*-----*/

Build : _481094f-20260618 OPENFOAM=2606 version=2606
Arch  : "LSB;label=32;scalar=64"
Exec  : simpleFoam
Date  : Sep 24 2026
Time  : 20:04:01
Host  : ODU-JRZWDF3
PID   : 3932
I/O   : uncollated
Case  : /home/s3kim/OpenFOAM/s3kim-v2606/run/pitzDaily
nProcs : 1
trapFpe: Floating point exception trapping enabled (FOAM_SIGFPE).
memory pool : not available
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationS

--> FOAM Warning : allowSystemOperations : Allowing user-supplied system call operations.
This can be a security risk if running untrusted cases
through e.g. 'coded' on-the-fly-compilation functionality.
```



OpenFOAM - ParaView

You can view 3D OpenFOAM simulation results in WSL using **either native Linux GUI rendering (WSLg)** or by **running ParaView directly on Windows** using your WSL files.

Option 1: Native WSLg Setup (Recommended for Windows 11)

On Windows 10 (build 19044+) and Windows 11, WSL includes WSLg, which automatically handles GUI displays without requiring extra configuration.

1. Install ParaView in Ubuntu:

```
sudo apt update  
sudo apt install -y paraview
```

2. Navigate to Your Case Directory:

```
cd $FOAM_RUN/pitzDaily
```

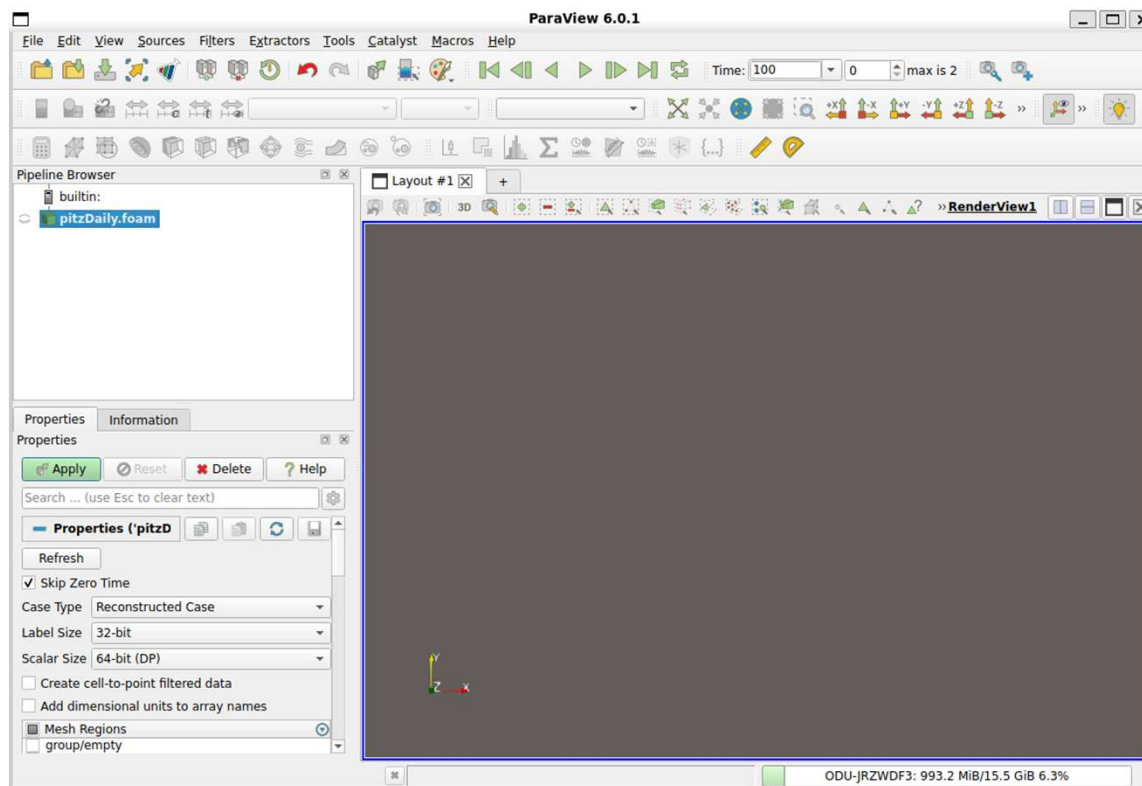
3. Launch ParaView using the native OpenFOAM utility command:

```
paraFoam
```



OpenFOAM - ParaView

Option 1: Native WSLg Setup (continued)



Note: In my test, WSLg hardware OpenGL rendering failed to show the CFD domain.

This is a common issue with ParaView running inside WSLg due to GPU acceleration handoff bugs between Windows and Linux Mesa drivers.

**Use Native Windows ParaView
(See Next Page)**



OpenFOAM - ParaView

Option 2: Windows Native ParaView (Best Performance Option)

If graphics rendering in WSL runs slow, installing ParaView on Windows provides direct access to your GPU hardware acceleration.

1. Install ParaView on Windows

Download and install the Windows version from [paraview.org](https://www.paraview.org).

2. Create the OpenFOAM Dummy File in WSL

In your Ubuntu terminal, navigate to your case directory and create a dummy .foam file:

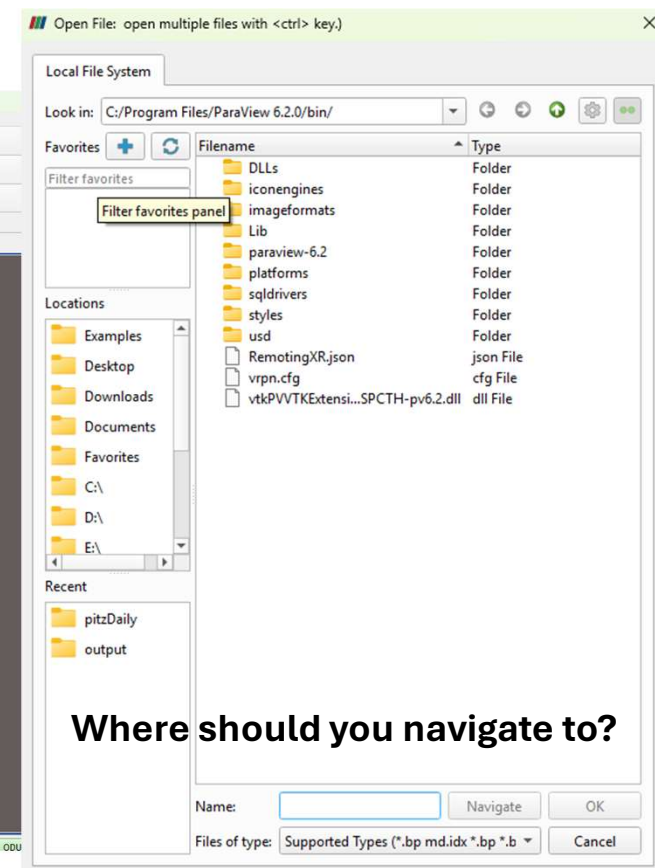
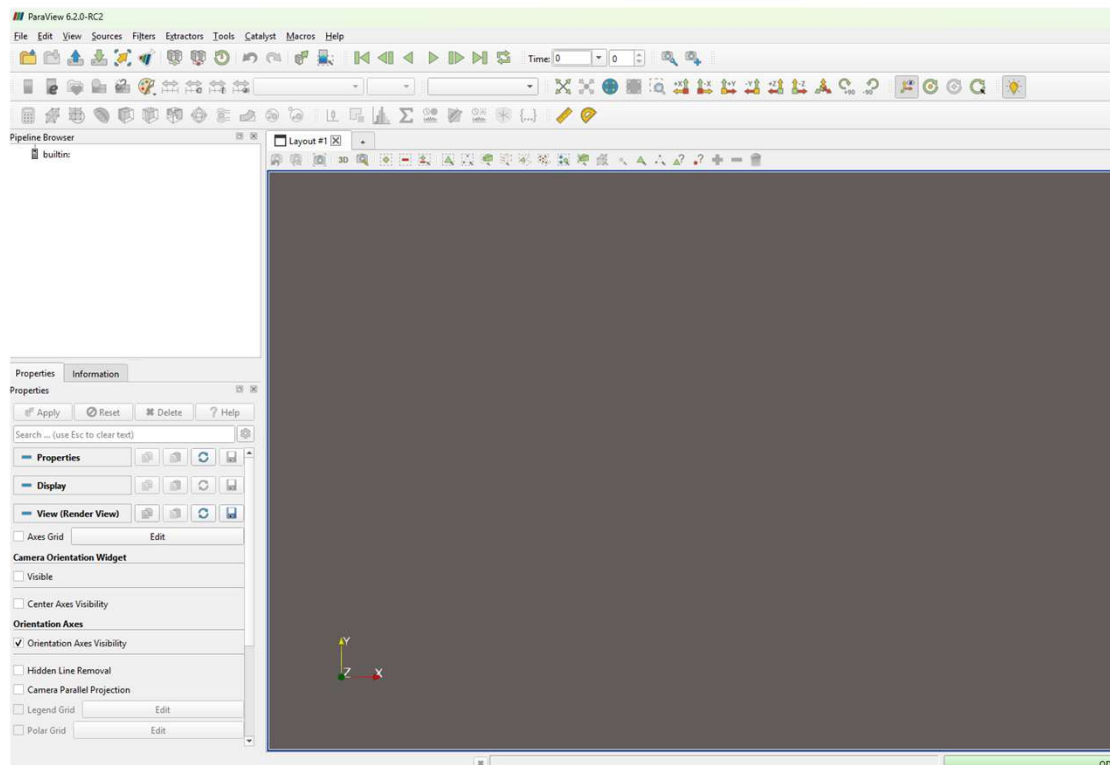
```
cd $FOAM_RUN/pitzDaily  
touch case.foam
```

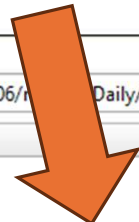
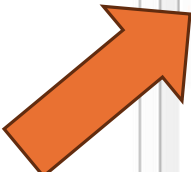


OpenFOAM - ParaView

3. Open the .foam File in Windows ParaView

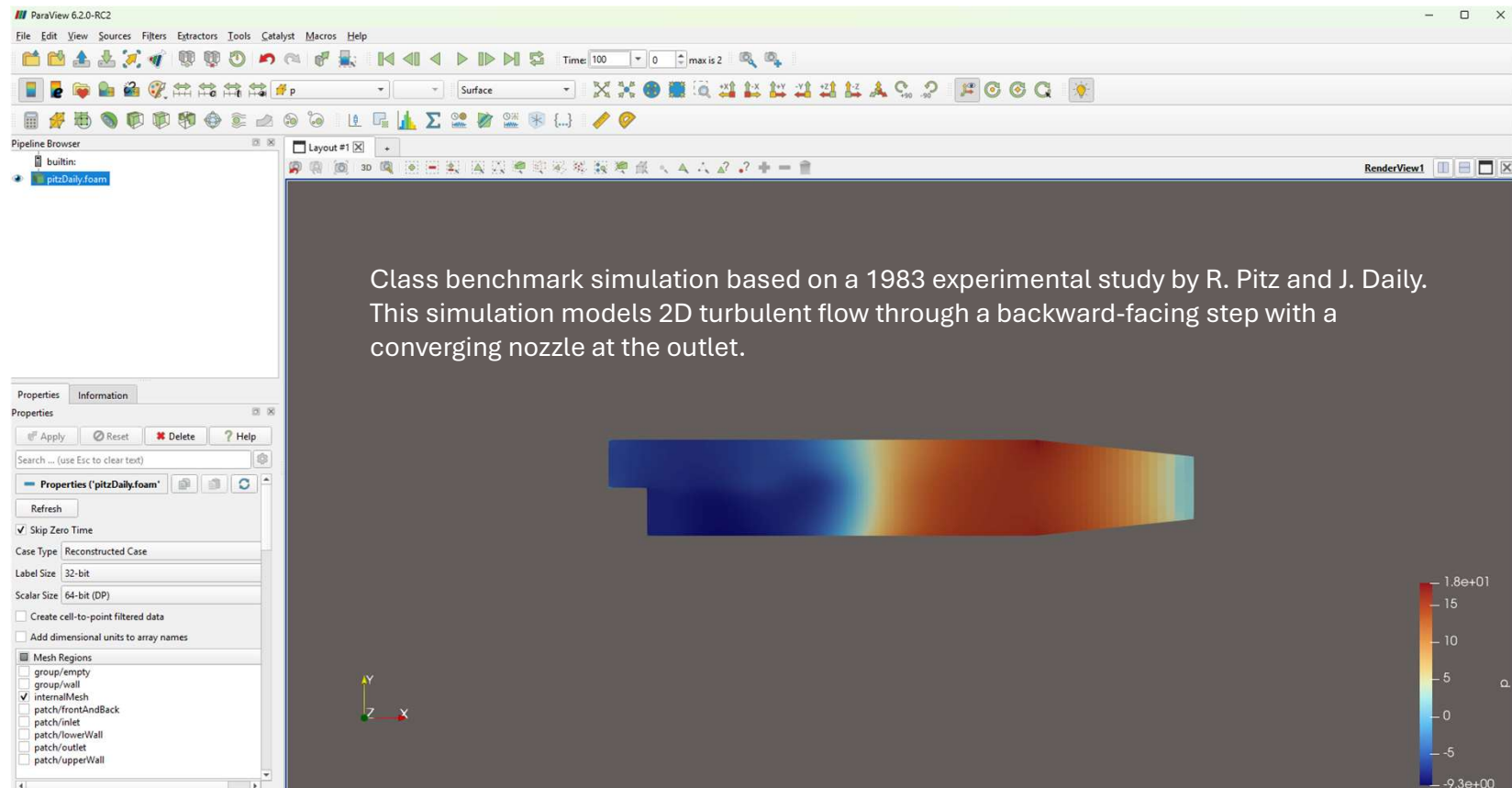
- Launch ParaView on Windows







OpenFOAM - ParaView





OpenFOAM - ParaView

Let's clean the case directory and rerun the simulation!

1. Run foamCleanTutorials:

This official OpenFOAM cleanup command safely removes all generated time directories, post-processing outputs, log files, and temporary .foam files while keeping your initial setup files (0/, constant/, and system/) intact.

`foamCleanTutorials`

2. Rerun the Simulation from Scratch:

Once cleaned, rebuild the mesh and launch the solver:

`blockMesh`
`simpleFoam`

```
s3kim@ODU-JRZWDF3:~/OpenFOAM/s3kim-v2606/run/pitzDaily$ ll
total 36
drwxr-xr-x 9 s3kim s3kim 4096 Sep 24 20:25 ./
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:03 ../
drwxr-xr-x 2 s3kim s3kim 4096 Sep 24 20:03 0/
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:04 100/
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:04 200/
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:04 281/
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:03 constant/
-rw-r--r-- 1 s3kim s3kim    0 Sep 24 20:25 pitzDaily.foam
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:04 postProcessing/
drwxr-xr-x 2 s3kim s3kim 4096 Sep 24 20:03 system/
s3kim@ODU-JRZWDF3:~/OpenFOAM/s3kim-v2606/run/pitzDaily$ foamCleanTutorials
Cleaning case /home/s3kim/OpenFOAM/s3kim-v2606/run/pitzDaily
s3kim@ODU-JRZWDF3:~/OpenFOAM/s3kim-v2606/run/pitzDaily$ ll
total 20
drwxr-xr-x 5 s3kim s3kim 4096 Sep 24 20:58 ./
drwxr-xr-x 3 s3kim s3kim 4096 Sep 24 20:03 ../
drwxr-xr-x 2 s3kim s3kim 4096 Sep 24 20:03 0/
drwxr-xr-x 2 s3kim s3kim 4096 Sep 24 20:58 constant/
drwxr-xr-x 2 s3kim s3kim 4096 Sep 24 20:03 system/
s3kim@ODU-JRZWDF3:~/OpenFOAM/s3kim-v2606/run/pitzDaily$ |
```



OpenFOAM - blockMesh

In OpenFOAM, blockMesh is the built-in, parametric mesh generation utility used to build structured hexahedral (3D block-based) grids.

When you type `blockMesh` in your terminal, the command reads an instruction file called `blockMeshDict`. It then outputs the actual 3D computational mesh files into the `constant/polyMesh/` folder (including `points`, `faces`, `owner`, `neighbor`, and `boundary`).

```
s3kim@ODU-JRZWF3:~/OpenFOAM/s3kim-v2606/run/pitzDaily$ blockMesh
/*-----*
|=====|
| \      | F ield      | OpenFOAM: The Open Source CFD Toolbox
|  \     | O peration  | Version: 2606
|   \    | A nd        | Website: www.openfoam.com
|    \   | M anipulation|
|=====|
/*-----*/
Build : _481094f-20260618 OPENFOAM=2606 version=2606
Arch : "x86_64-linux-gcc"
Create time

Creating block mesh from "system/blockMeshDict"
Creating block edges
No non-planar block faces defined
Creating topology blocks

Creating topology patches - from boundary section

Creating block mesh topology - scaling/transform applied later
```



OpenFOAM - blockMesh

```
/*----- C++ -----*/
FoamFile
{
    version      2.0;
    format       ascii;
    class        dictionary;
    object       blockMeshDict;
}
// *****

scale 0.001;

vertices
(
    (-20.6 0 -0.5)
    (-20.6 25.4 -0.5)
    (0 -25.4 -0.5)
    (0 0 -0.5)
    (0 25.4 -0.5)
    (206 -25.4 -0.5)
    (206 0 -0.5)
    (206 25.4 -0.5)
    (290 -16.6 -0.5)
    (290 0 -0.5)
    (290 16.6 -0.5)

    (-20.6 0 0.5)
    (-20.6 25.4 0.5)
    (0 -25.4 0.5)
    (0 0 0.5)
    (0 25.4 0.5)
    (206 -25.4 0.5)
    (206 0 0.5)
    (206 25.4 0.5)
    (290 -16.6 0.5)
    (290 0 0.5)
    (290 16.6 0.5)
);

negY
(
    (2 4 1)
    (1 3 0.3)
);
```

```
blocks
(
    hex (0 3 4 1 11 14 15 12)
      (18 30 1)
      simpleGrading (0.5 $posY 1)

    hex (2 5 6 3 13 16 17 14)
      (180 27 1)
      edgeGrading (4 4 4 4 $negY 1 1 $negY 1 1 1 1)

    hex (3 6 7 4 14 17 18 15)
      (180 30 1)
      edgeGrading (4 4 4 4 $posY $posYR $posYR $posY 1 1 1 1)

    hex (5 8 9 6 16 19 20 17)
      (25 27 1)
      simpleGrading (2.5 1 1)

    hex (6 9 10 7 17 20 21 18)
      (25 30 1)
      simpleGrading (2.5 $posYR 1)
);

edges
(
);

boundary
(
    inlet
    {
        type patch;
        faces
        (
            (0 1 12 11)
        );
    }
    outlet
    {
        type patch;
        faces
        (
            (8 9 20 19)
            (9 10 21 20)
        );
    }
    upperWall
    {
        type wall;
    }
);
```



OpenFOAM - blockMesh

The control file (`system/blockMeshDict`) specifies how the 3D space is built:

- **scale / convertToMeters:** Multiplies coordinate dimensions (e.g., 0.001 scales millimeters to meters).
- **vertices:** Defines spatial 3D points (X, Y, Z) marking the corners of blocks.
- **edges:** Defines curvature (such as arcs or splines) between vertices if the mesh isn't made up of flat straight lines.
- **blocks:** Groups 8 vertices together to form a 3D volume (hexahedron), setting the cell resolution (e.g., $100 \times 20 \times 1$) and cell expansion/grading ratios.
- **boundary:** Assigns surface faces of the blocks to physical boundary patches (like inlet, outlet, walls, or empty for 2D cases).